

Formula sheet for Quiz 1

Note to students: you do not need to print or memorize this. You will be handed a copy with the quiz. It lists more formulas than any one question needs, so part of each question is deciding which of these applies.

Models

linear, one input	$\hat{y} = w_0 + w_1 x$
linear, many inputs	$\hat{y} = w_0 + \sum_{j=1}^d w_j x_j$
logistic	$\hat{p} = \sigma \left(w_0 + \sum_j w_j x_j \right), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$

Losses

residual for example i	$r_i = y_i - \hat{y}_i$
mean squared error	$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$
sum of squared errors	$\text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (\text{note: no division by } n)$
mean absolute error	$\text{MAE} = \frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $
binary cross-entropy	$\text{BCE} = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]$

Optimization

gradient	$\nabla \mathcal{L}(\mathbf{w}) = \left(\frac{\partial \mathcal{L}}{\partial w_0}, \frac{\partial \mathcal{L}}{\partial w_1}, \dots, \frac{\partial \mathcal{L}}{\partial w_N} \right)$
gradient descent, all parameters	$\mathbf{w}_{\text{new}} = \mathbf{w}_{\text{old}} - \lambda \nabla \mathcal{L}(\mathbf{w}_{\text{old}})$
gradient descent, one parameter j	$w_{\text{new}}^j = w_{\text{old}}^j - \lambda \frac{\partial \mathcal{L}}{\partial w^j}$
at a minimum	$\nabla \mathcal{L}(\mathbf{w}) = 0$

Reference

noise floor for $y = f(x) + \varepsilon$	$\min \text{MSE} = \text{Var}(\varepsilon) = \sigma_\varepsilon^2$
majority-class baseline	$\max(\bar{y}, 1 - \bar{y})$